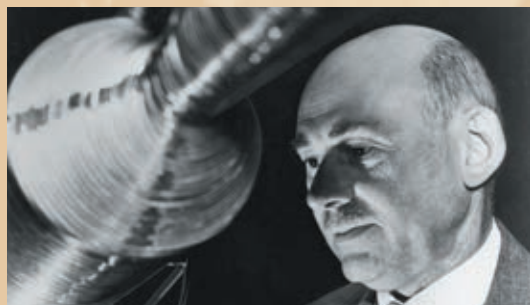


ROBERT GODDARD

BORN IN WORCESTER, MASSACHUSETTS, Robert Goddard (1882–1945) was inspired by reading science fiction as a youth to a lifelong obsession with space travel. He pursued research into rocket motors as a student and then professor at Clark University, and in 1926 launched the first rocket fueled by a liquid propellant. Goddard's work drew the attention of Lindbergh, and through him funding from the Guggenheim Foundation. This allowed Goddard to experiment with bigger and faster rockets in the 1930s, working at Roswell, New Mexico. He also developed sophisticated control systems.



SOLITARY GENIUS

By nature a loner, Robert Goddard mostly worked in isolation, keeping details of his experiments secret. As a result, the practical development of rocketry in Germany occurred without reference to his pioneering work.

By the 1920s the theoretical possibility of sending rockets into space had been established, but it was still a topic more likely to capture the imagination of oddball enthusiasts than mainstream scientists. Goddard, Oberth, and the aging Tsiolkovsky – elevated to membership of the Soviet Academy of Sciences by the new Bolshevik regime in 1919 – were revered by groups of youths who set up rocket clubs and societies dedicated to interplanetary travel. Space fiction flourished, from the frivolous adventures of Buck Rogers and Flash Gordon to the fantastic prophesies of Fritz Lang's 1929 movie *Frau im Mond* (Woman in the Moon). But the association of rockets with fantasy fiction and utopian projects for space colonies deterred more level-headed people from taking the subject seriously.

Still, practical progress was made, initially with Goddard in the lead. His first liquid-propelled rocket, launched in 1926, reached only a modest height of 41ft

(12.5m), but by 1930 he fired a rocket up to 2,000ft (600m), and in 1935 one of his rockets rose to 7,500ft (2,300m).

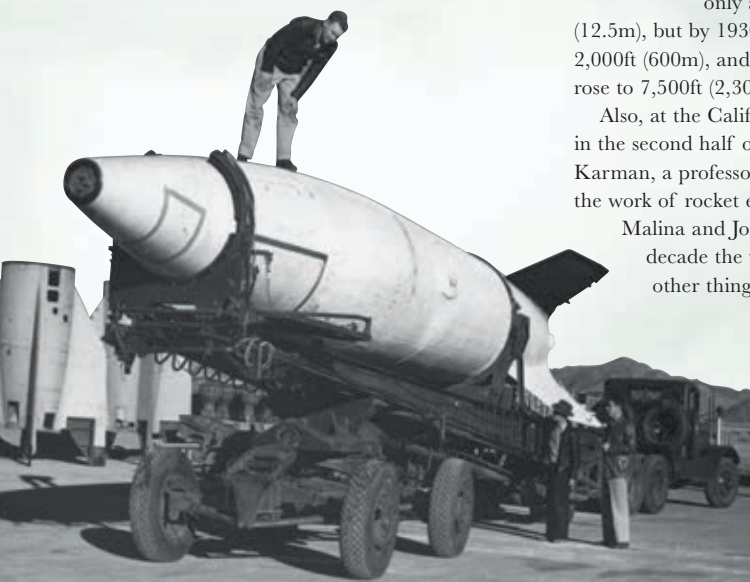
Also, at the California Institute of Technology in the second half of the 1930s, Theodore von Karman, a professor of aeronautics, encouraged the work of rocket experimenters such as Frank Malina and John Parsons. In the following decade the work at Caltech led to, among other things, development of rocket-

assisted takeoff for aircraft. Another American group traced a remarkable path from fantasy to reality: the American Interplanetary Society, founded in 1930 mostly by science fiction writers, turned into the American Rocket Society in 1934, dedicated to practical experimentation, and in the 1940s gave birth to Reaction Motors Inc., the company that built the engine that powered the Bell X-1.

Military interest

The US was far from leading the world in rocket development. Through the 1920s and 1930s, it was in the Soviet Union and above all Germany that the idea of space exploration excited most serious interest, and that military authorities came to view rocket weapons as potential war-winners.

The initial development of liquid-propellant rockets in Germany was largely the work of the Verein für Raumschiffahrt (Society for Space Travel). In 1931–32, at their Raketenflugplatz (rocket flying field) in a Berlin suburb, the German enthusiasts carried out around 100 rocket launches, reaching altitudes up to 4,922ft (1,552m). Their work was being monitored by the German army, dreaming of escape from the severe restrictions placed upon it by the Versailles peace treaty. Suspicious of what they saw as the “humbugs, charlatans, and scientific cranks” of the rocket group, the army decided to recruit them into its own weapons program, where they could be kept under supervision. The only taker



CAPTURED ROCKET

Wernher von Braun's V-2 rocket was the starting point for both the American and Soviet space and missile programs after World War II. This one was brought to the United States.

